Loan Default Prediction Project – Report

1. INTRODUCTION AND PROJECT OBJECTIVE

In today's financial landscape, lending institutions rely heavily on data to assess the risk of loan applicants. Defaulting on loans can significantly impact a bank's profitability and stability. As a result, credit risk modeling has become a key area where machine learning adds real-world value. By analyzing historical loan records and applicant characteristics, predictive models can help identify high-risk individuals before loans are approved.

This project applies supervised machine learning to predict whether a loan applicant is likely to default, using a dataset from Kaggle. The goal is to build and compare classification models (Random Forest and Logistic Regression), and to deploy a functional prediction interface. We build a classification model that classifies applicants into two categories:

0 – Will not default

1 – Will default

The pipeline includes data cleaning, preprocessing, training models (Random Forest and Logistic Regression), evaluating their performance and deploying a prediction function for new applicants.

The main objectives of this project are to:

- Explore and understand the structure and insights from the Loan Default dataset.
- Preprocess and transform the data into a format suitable for machine learning.
- Train and compare classification models (Random Forest & Logistic Regression).
- Evaluate models using precision, recall, F1-score, and confusion matrix.
- Deploy a prediction interface for real-time risk classification.
- Interpret model results and feature importance to offer real-world insights.

Data dictionary

- Column: Name Description
- LoanID: Unique identifier for each loan record
- Age: Age of the applicant
- Income: Annual income of the applicant (in local currency e.g., GHS)
- LoanAmount: Total loan amount requested
- CreditScore: Applicant's credit score (300–850 scale)

- MonthsEmployed: Total months the applicant has been employed
- NumCreditLines: Number of active credit lines (loans, credit cards, etc.)
- InterestRate: Annual interest rate (%) on the loan
- LoanTerm: Duration of loan repayment (in months, e.g., 12/24/36/60)
- DTIRatio: Debt-to-Income ratio (e.g., 0.5 means 50% of income goes to debt)
- Education: Highest education level (e.g., High School, Bachelor's, Master's)
- EmploymentType: Employment status (e.g., Full-time, Part-time, Unemployed)
- MaritalStatus: Marital status (e.g., Single, Married, Divorced)
- HasMortgage: Whether the applicant has an existing mortgage (Yes/No)
- HasDependents: Whether the applicant has dependents (children or others) (Yes/No)
- LoanPurpose: Reason for the loan (e.g., Auto, Education, Business, Personal, Other)
- HasCoSigner: Whether the applicant has a co-signer or guarantor (Yes/No)
- Default Target variable: 1 if the applicant defaulted, 0 otherwise

2. DATA IMPORT AND OVERVIEW

- The dataset consists of 255,347 rows and 18 columns with no missing values. The datasets comprise of both numeric and categorical datatypes. Applicants range from 16 to 69 years, with an average of 43. A box plot graph was generated to check outliers, the histogram chart showed variable distributions.

  Numerical Analysis

  - The annual income of applicants ranges widely from Annual earnings range widely from $15,000 to $149,999, with an average income of $82,499. A standard deviation of $38.96 indicates a high considerable variation in income levels across the sample.
  - Loan amount varies between $5,000 and $249,999 with an average of $127,579. A standard deviation of $70840.71
  - The highest credit score to enable Scores range from 300 to 849 with a mean of 574. The standard deviation shows a wide credit spectrum.

Categorical Analysis

- Education & Employment Categories: Perfect 25% distribution across all levels provides equal representation for analyzing how educational attainment and work status influence default risk.
- Marital Status: Balanced three-way split (33%) ensures sufficient sample sizes for comparing loan performance across different relationship statuses.
- Mortgage/Dependents/CoSigner: The exact 50/50 distributions across these yes/no variables suggest half of the borrowers have or do not have mortgages, dependents or cosigner.
- Loan Purpose: Shows diverse lending purposes i.e. home, business, education, and other needs.

Data Visualization: Explored the distribution of numeric features via histograms and boxplots to detect outliers and analyzed categorical columns through pie charts to understand class proportions.

Data Preprocessing: In the data cleaning process, no missing values were found. Loan1D was removed because it can't be used for thorough analysis. All categorical data were replaced with 1s and 0s using the get dummies function. The values were standardized using the StandardScaler to ensure all values has a mean of 0 and a standard deviation of 1, this will ensure that all features have the same footing. A correlation heatmap was generated to study variable relationships. The binary target variable was retained("Default').

- Model Development: Built and trained two classification models: a Random Forest Classifier with 50 trees and a maximum depth of 10, and a Logistic Regression model with up to 1000 iterations.
- Model Evaluation: Assessed both models on test data using classification reports (precision, recall, F1-score) and confusion matrices. Visualized model performance differences to inform selection. However, due to an imbalanced dataset in the default column, smoting was done to balance the dataset. Comparison of classification reports before and after was done.

Below are the various classification reports gotten:

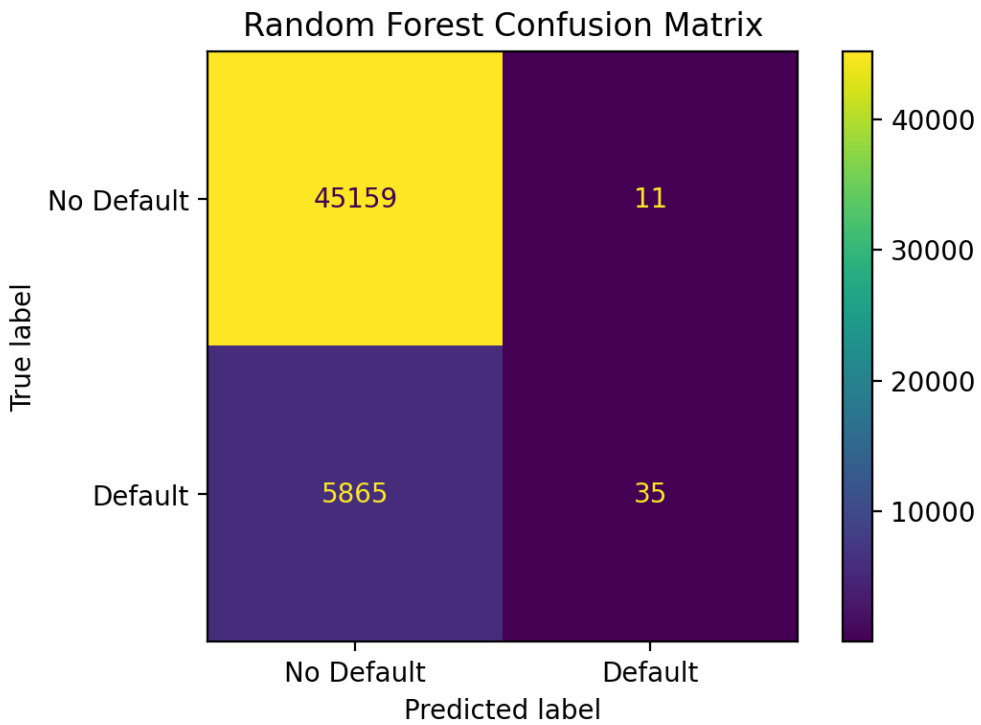
CLASSIFICATION AND CONFUSION MATRIX REPORTS BEFORE SMOTING

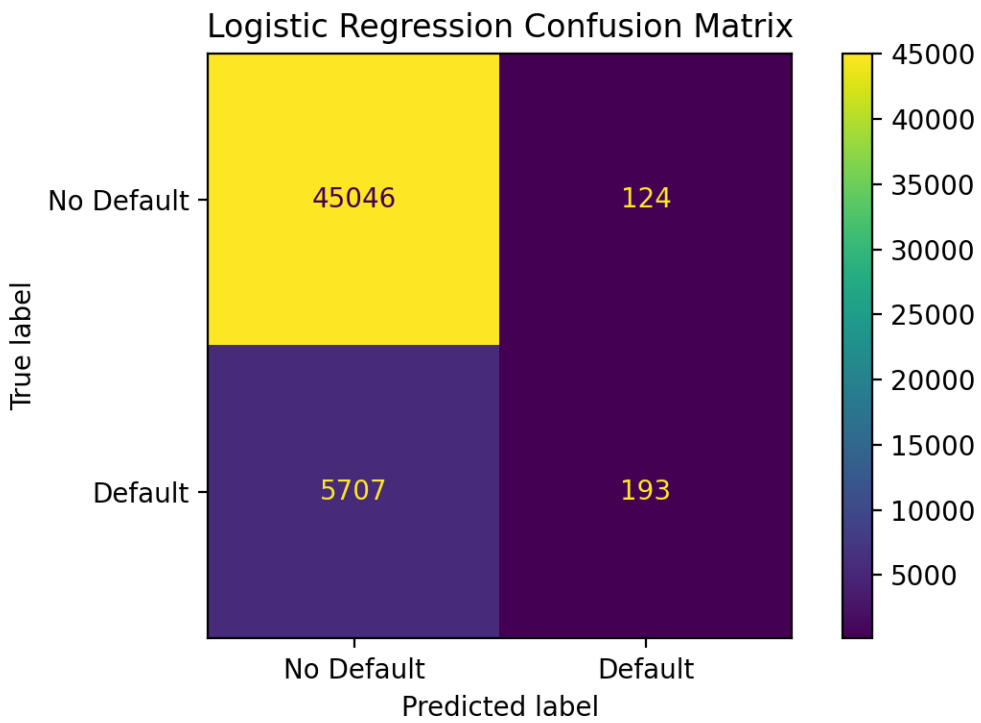
- **Random Forest Report**

	precision	recall	f1-score	support
0	0.885	0.9998	0.9389	45,170
1	0.7556	0.0058	0.0114	5,900
accuracy	0.8849	0.8849	0.8849	0.8849
macro avg	0.8203	0.5028	0.4752	51,070
weighted avg	0.8701	0.8849	0.8318	51,070

- **Logistic Regression Report**

	precision	recall	f1-score	support
0	0.8876	0.9973	0.9392	45,170
1	0.6088	0.0327	0.0621	5,900
accuracy	0.8858	0.8858	0.8858	0.8858
macro avg	0.7482	0.515	0.5006	51,070
weighted avg	0.8554	0.8858	0.8379	51,070


Random Forest Confusion Matrix
No Default
Default
True label
45159
11
5865
35
No Default
Default
Predicted label
40000
30000
20000
10000


Logistic Regression Confusion Matrix
No Default
Default
True label
45046
124
5707
193
No Default
Default
Predicted label
45000
40000
35000
30000
25000
20000
15000
10000
5000

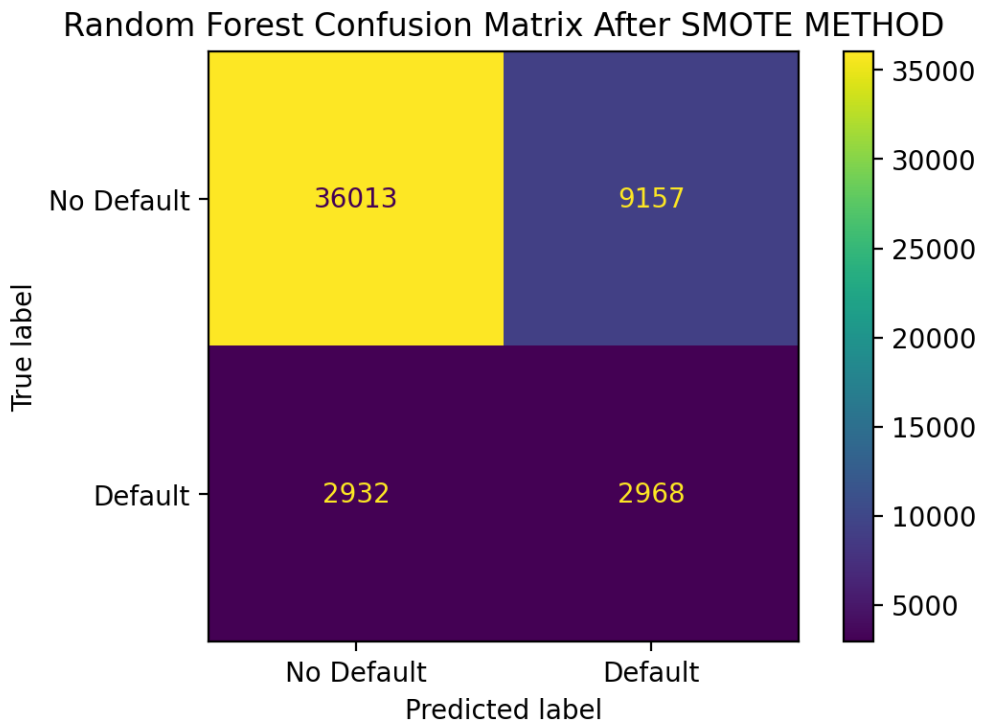
CLASSIFICATION AND CONFUSION MATRIX REPORTS AFTER SMOTING

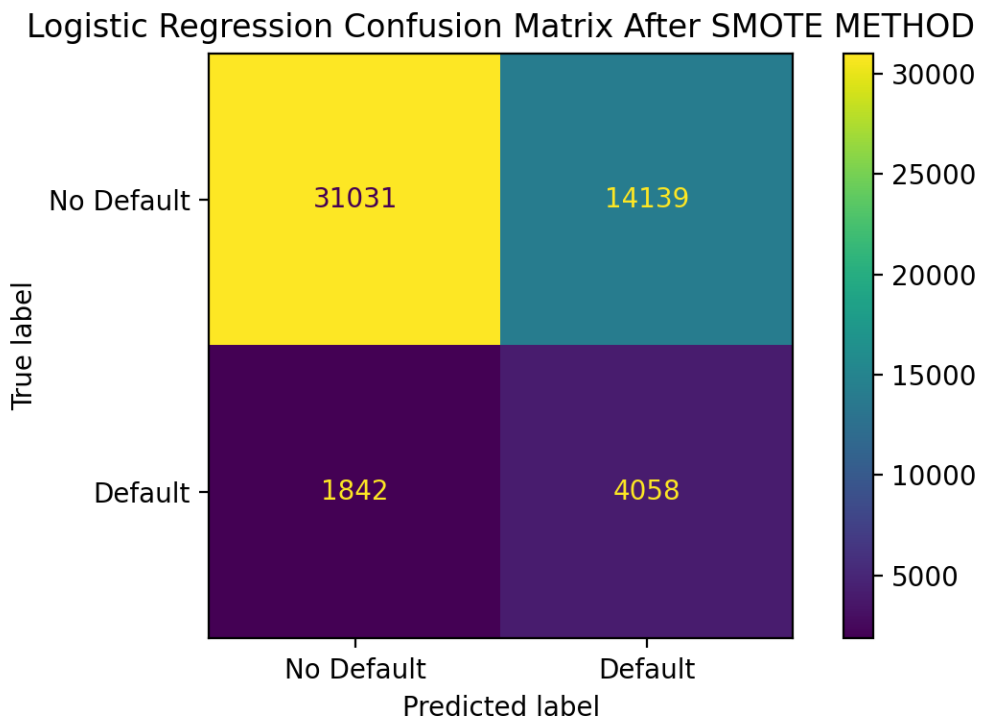
Random Forest Report

	precision	recall	f1-score	support
0	0.9247	0.7973	0.8563	45,170
1	0.2448	0.5031	0.3293	5,900
accuracy	0.7633	0.7633	0.7633	0.7633
macro avg	0.5847	0.6502	0.5928	51,070
weighted avg	0.8462	0.7633	0.7954	51,070

Logistic Regression Report

	precision	recall	f1-score	support
0	0.944	0.687	0.7952	45,170
1	0.223	0.6878	0.3368	5,900
accuracy	0.6871	0.6871	0.6871	0.6871
macro avg	0.5835	0.6874	0.566	51,070
weighted avg	0.8607	0.6871	0.7423	51,070


Random Forest Confusion Matrix After SMOTE METHOD
No Default
Default
True label
36013
9157
2932
2968
No Default
Default
Predicted label
35000
30000
25000
20000
15000
10000
5000


Logistic Regression Confusion Matrix After SMOTE METHOD
No Default
Default
True label
31031
14139
1842
4058
No Default
Default
Predicted label
30000
25000
20000
15000
10000
5000

COMPARISON/ EFFECT OF SMOTING

MODEL/DEFAULT OR NOT	STAGE	PRECISION	RECALL	F1-SCORE	ACCURACY
Random Forest (0)	Before smoting	0.88	0.99	0.93	0.88
Random Forest(1)	Before smoting	0.76	0.006	0.012	0.88
Random Forest(0)	After smoting	0.92	0.79	0.85	0.76
Random Forest(1)	After smoting	0.24	0.50	0.76	0.88
Logistic (0) Regression	Before smoting	0.88	0.99	0.93	0.88
Logistic (1) Regression	Before smoting	0.60	0.03	0.06	0.88
Logistic (0) Regression	After smoting	0.94	0.68	0.79	0.68
Logistic (1) Regression	After smoting	0.22	0.68	0.33	0.68

Interpretation of Model Results Before and After Applying SMOTE

This table compares how two models — Random Forest and Logistic Regression — perform before and after applying SMOTE (Synthetic Minority Oversampling Technique) on an imbalanced dataset with two classes:

- 0 = Not Defaulted (Majority class)
- 1 = Defaulted (Minority class)

We're assessing each model using Precision, Recall, F1-Score, and Accuracy.

BEFORE SMOTING (on the original imbalanced dataset)

🔷 Random Forest

- Class 0 (Not Default):
  - Precision: 0.88 — 88% of predicted "not default" cases were correct.
  - Recall: 0.99 — Almost all actual "not default" cases were captured.

  - F1-score: 0.93 — High balance between precision and recall.
- Class 1 (Default):
  - Precision: 0.76 — Looks decent, but...
  - Recall: 0.006 — Almost none of the actual defaulters were identified.
  - F1-score: 0.012 — Model is practically useless for detecting defaulters.
- Accuracy: 0.88 — Looks high but misleading, because it's biased toward predicting the majority class.

🔹 Logistic Regression

- Class 0 (Not Default):
  - Precision: 0.88, Recall: 0.99, F1-score: 0.93 — just like Random Forest.
- Class 1 (Default):
  - Precision: 0.60, Recall: 0.03, F1-score: 0.06 — barely recognizes defaulters.
- Accuracy: 0.88 — Again, looks okay but hides poor performance on the minority class.

Key Insight: Before SMOTE, both models focused heavily on the majority class. The models were almost blind to defaulters. Accuracy was inflated due to class imbalance.

AFTER SMOTING (balancing the dataset)

🔹 Random Forest

- Class 0 (Not Default):
  - Precision: 0.92 — Slight drop from before.
  - Recall: 0.79 — Model is now misclassifying some non-defaulters.
  - F1-score: 0.85 — Still solid.
- Class 1 (Default):
  - Precision: 0.24 — Many false positives (predicts default when it's not).
  - Recall: 0.50 — Big improvement; the model now catches half of all true defaulters.
  - F1-score: 0.76 — Major improvement from before.
- Accuracy: 0.88 — Same as before, but now it reflects more balanced performance.

🔹 Logistic Regression

- Class 1 (Default):
  - Precision: 0.22 — Low, but recall improves dramatically to 0.68.
  - F1-score: 0.33 — A significant jump from 0.06 before.
- Accuracy: 0.68 — Lower overall accuracy, but better fairness in predictions.

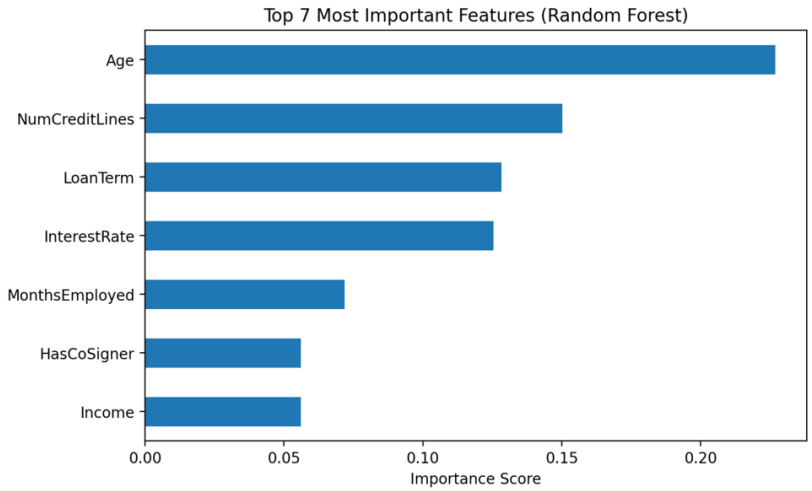
Key Insight: After SMOTE, the models became much better at detecting defaulters (minority class). Even though precision dropped slightly, recall and F1-score increased significantly, which is critical when catching high-risk individuals.

Final Summary

Scenario	Class 1 (Default) Performance	Overall Accuracy	Useful?
Before SMOTE	Very poor (almost ignored)	High (misleading)	No
After SMOTE	Much better	Moderate	Yes

- Accuracy is not reliable when data is imbalanced. Focus on recall and F1-score for the minority class.
- SMOTE improves the model's ability to detect defaulters by generating synthetic examples of the underrepresented class.
- Random Forest performs better than Logistic Regression, especially after SMOTE.

- Feature Importance: Identified the top 7 most significant predictors for loan default using the feature importance from the Random Forest model after balancing the dataset.



- Prediction Interface: Developed a Streamlit web app interface allowing user-friendly input of applicant data to obtain real-time loan default risk predictions. A sample decision tree visualization from the Random Forest provides transparency into model decisions.

Key Findings

- The Random Forest model generally offers superior performance on classification metrics compared to Logistic Regression, suggesting better handling of complex feature interactions in the dataset.
- Top features impacting default prediction includes:

    - Age
    - Whether the person has a co-signer or not
    - Loan Term
    - Interest Rate
    - Months Employed
    - Income
    - Number of Credit Lines
- The model effectively differentiates between applicants likely to default and those who are not, allowing financial institutions to assess risk proactively.

Practical Implications

Financial institutions can leverage this predictive model to:

- Improve loan approval decisions: By identifying high-risk applicants early, reducing non-performing loans.
- Optimize interest rates and terms: Tailoring offers based on predicted default probability.
- Support regulatory compliance and risk management: Providing data-driven justification for lending policies.
- Enhance customer targeting: Identifying low-risk customers for better loan products and higher profits.

Limitations and Recommendations

- Data Quality: The model's accuracy depends heavily on data accuracy and completeness. Real-world applications should ensure continual data validation.
- Model Complexity vs Interpretability: While Random Forest offers strong predictive power, logistic regression provides clearer interpretability. A hybrid approach or explainer tools (like SHAP values) could further enhance trust.
- Feature Coverage: Additional relevant features such as credit history length, asset details, or macroeconomic indicators might improve predictions.
- Deployment and Monitoring: Continuous monitoring of model performance over time is essential to detect performance drift due to changing borrower behaviors or economic conditions.

Future Work

- Incorporate additional data sources to enrich applicant profiles.
- Explore advanced models (e.g., gradient boosting, neural networks) for improved accuracy.
- Implement explainability techniques for better stakeholder understanding.
- Build an end-to-end automated pipeline integrating data ingestion, retraining, and deployment.

Conclusion

This project demonstrates the practical utility of machine learning in managing credit risk. By combining rigorous data preprocessing, careful model training, evaluation, and an interactive prediction tool, the work equips lending institutions with an effective method for mitigating loan default risk, improving decision-making, and ultimately enhancing financial stability.

LINKS FOR STREAMLIT APP

Local URL: http://localhost:8501

Network URL: http://192.168.100.122:8501